Nonlinear hydropower scheduling with coupled hydraulics and delayed river networks

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Abstract

Short-term hydropower scheduling must decide when to generate or pump and how much water to release. The water available to downstream plants depends on earlier releases and river travel times. Reservoir levels and shared tunnel losses also affect the power obtained from each unit. We formulate the schedule as a mixed-integer nonlinear program with unit commitment, finite gate choices and delayed river transport. Reservoir balances use an implicit discrete-gradient step that respects stored potential energy. Pressure losses and power conversion are nonlinear, and tabulated equipment curves keep their specified interpolants. For fixed river travel times, releases are shifted through the network without averaging at confluences. Distributed delays use interval cohorts and track water still in transit. SCIP searches the full model using feasible starts prepared by mixed-integer proposals and nonlinear dispatch. A separate integration checks the controls on the optimization grid and a finer grid. Optional spinning capacity uses feasible unit operating points at scheduled hydraulic node heads, with shared tailwater and qualified response limits. We state the limits of the resulting objective-gap certificate. The implementation uses Julia and open-source solvers.

1 The scheduling problem

A reservoir can generate now or save water for later. A release lowers its level and changes the energy available from the next release. In a cascade, the water may pass through several plants before leaving the system. The upstream schedule then determines when water becomes available downstream. Shared tunnels impose another dependency, since a plant's discharge affects pressure losses for other plants on the waterway. A unit may be stopped, but when running it must stay within its operating range. These discrete choices and the dependence of generation on head have long been studied together [1].

We take the network, initial volumes, operating histories, prices, inflows, equipment curves and operating limits as given. The optimizer chooses unit states, continuous unit discharges, outlet openings and optional spinning reserve allocations. It selects each adjustable tunnel position from a finite set. The associated volumes, flows, heads, river arrivals and electrical outputs must satisfy the equations below. We maximize net electrical revenue less operating penalties, with a terminal value for the water left in the system.

We assume deterministic inputs and quasi-steady tunnel hydraulics: pressures and flows are in equilibrium within each scheduling interval while reservoir storage evolves over time. Tunnel inertia and water hammer are excluded. Discharge and travel-time response curves describe the rivers, without solving river momentum equations during scheduling. We do not include stochastic decisions. Pumps use supplied hydraulic efficiency functions and constant motor efficiency; generating and pumping modes retain separate positive flow and power magnitudes. The reserve capacity model below does not simulate hydraulic activation across the watercourse.

The optimization uses nonlinear pressure-loss and power equations. Linear approximations provide initial discrete choices for the solution procedure in Section 8.

2 Water balance and hydraulic head

Let $0 = \tau_0 < \cdots < \tau_T$ be time edges in hours, $\Delta_t = \tau_{t+1} - \tau_t$ the interval duration, and $m_t = (\tau_t + \tau_{t+1})/2$ its midpoint, for $t = 0, \ldots, T-1$. Interval inputs are piecewise constant, with changes at

these edges. We record volumes at the edges and evaluate interval hydraulics using volume-averaged reservoir heads.

Hydraulic head is mechanical energy per unit weight, expressed in metres. At a reservoir it is the water-surface elevation. At a flooded junction it includes elevation and pressure divided by ρg, where ρ is water density and g is gravitational acceleration. Water flows down a head gradient. Friction dissipates energy along the waterway, and a turbine can extract power from the flow. The equipment relationships account for velocity corrections and local losses.

Volume is measured in Mm^3 (one million cubic metres, $10^6\mathrm{m}^3$), discharge in $\frac{\mathrm{m}^3}{\mathrm{s}}$ and time in hours. The conversion factors are

$$\kappa = \frac{3600}{10^6} = 0.0036, \quad \gamma = \frac{\rho g}{10^6} = 0.00981 \\ \left(\rho = 1000\frac{\mathrm{kg}}{\mathrm{m}^3}, \quad g = 9.81\frac{\mathrm{m}}{\mathrm{s}^2}\right). \tag{1}$$

Thus $\kappa\Delta q$ converts a discharge over an interval into a volume, and $\gamma q h$ is hydraulic power at discharge q and net head h. For example, a release of $10\frac{\mathrm{m}^3}{\mathrm{s}}$ for one hour removes $0.036\mathrm{Mm}^3$ of water. We use the same conversion for net inflow in the reservoir balance.

Let $\mathcal{R}$, $\mathcal{J}$ and $\mathcal{B}$ index reservoirs, pressure junctions and prescribed-head boundaries. A node n is any of these three. Tunnels, instantaneous channels, generating and pumping units and delayed river reaches have index sets $\mathcal{E}$, $\mathcal{C}$, $\mathcal{G}$ and $\mathcal{D}$, respectively. The units at plant p form $\mathcal{G}_p$. Table 1 lists the variables. Underlines and overlines denote supplied lower and upper bounds.

Table 1: Variables and units. Storage and river inventory are recorded at time vertices. The other indexed quantities refer to control intervals.

Variable	Meaning	Unit
V_{it}, W_{rt}	Reservoir volume, river water in transit	Mm^3
H_{nt}, h_{pt}	Node head, effective generating head or pumping lift	m
Q_{et}, X_{ct}, q_{gt}	Tunnel, channel, unit discharge	$\frac{\mathrm{m}^3}{\mathrm{s}}$
R_{rt}, d_{rt}	Reach entry flow, reservoir withdrawal	$\frac{\mathrm{m}^3}{\mathrm{s}}$
A_{rt}	Volume arriving from a delayed reach	Mm^3
P_{gt}	Positive production or consumption magnitude	MW
$r^{\pm}_{gkt}, \sigma^{\pm}_{kt}$	Spinning allocation, group shortfall	MW
u_{gt}, U_{pt}	Unit and plant operating states	$\{0,1\}$
b_{et}, a_{rt}	Tunnel position, reservoir outlet opening	$[0,1]$

Let F_{nt} be net flow into node n from the modeled connections, excluding natural reservoir inflow I_{it}. The matrices B^E, B^C and B^G record the tunnel, channel and unit connections. For a connection between nodes, its column is -1 at the source, $+1$ at the destination, and zero elsewhere. Negative tunnel/channel discharge reverses these contributions. Unit endpoints always follow their actual water direction, so a pump withdrawing from the lower reservoir and delivering to the upper one uses nonnegative q. Arrival volumes from rivers and withdrawals into them give the remaining terms:

$$F_{nt} = \sum_e B^E_{ne} Q_{et} + \sum_c B^C_{nc} X_{ct} + \sum_g B^G_{ng} q_{gt} + \sum_r D_{nr}\frac{A_{rt}}{\kappa\Delta_t} - \sum_r S_{nr} d_{rt}. \tag{2}$$

$D_{nr} = 1$ if reach r delivers into node n, and $S_{nr} = 1$ if an outlet at n withdraws water into that reach. The other entries are zero. Dividing A_{rt} by $\kappa\Delta_t$ converts arrival volume to flow rate. For a tunnel or plant outfall, F includes the withdrawal from the source node. The outfall enters the river balance below. Its outlet head can depend on a downstream reservoir even though the water reaches that reservoir later (Figure 1).

Water stored at the next time edge equals the present volume plus net inflow over the interval. A pressure junction stores no water, so its net inflow must be zero:

$$V_{i,t+1} - V_{it} = \kappa\Delta_t(I_{it} + F_{it}), \quad V_{i0} = V_i^{\mathrm{init}}. \tag{3}$$

$$F_{jt} = 0 \quad (j \in \mathcal{J}), \quad \underline{V}_{it} \le V_{it} \le \overline{V}_{it} \quad (t = 0, \ldots, T). \tag{4}$$

Initial volumes and volume limits are given data. Junction balances determine how flows meet and divide. Boundary nodes have prescribed heads and may exchange water with the exterior. Let $Z_i(V)$ be reservoir surface elevation and Ψ_i an antiderivative of Z_i. The interval head is the volume average of that elevation:

$$\overline{Z}_i(a,b) = \begin{cases} \frac{\Psi_i(b)-\Psi_i(a)}{b-a} & \text{if } a \neq b \\ Z_i(a) & \text{if } a = b \end{cases}, \quad H_{it} = \overline{Z}_i\big(V_{it}, V_{i,t+1}\big). \tag{5}$$

$$\underline{H}_j \le H_{jt} \le \overline{H}_j, \quad H_{bt} = H_b^{\text{given}}. \tag{6}$$

Z_i converts volume to surface elevation. It is an increasing supplied table, or $Z_i(V) = z_i + s_i V + c_i V^2$ with given coefficients. The ending volume is unknown in (3) and also enters (5). We solve these coupled equations implicitly. The head is a discrete gradient of stored potential [2]: multiplying it by the interval volume change gives exactly $\Psi_i(b) - \Psi_i(a)$. In MWh, the gravitational energy change is $\big(\frac{0.00981}{\kappa}\big)(\Psi_i(b) - \Psi_i(a))$. A common elevation datum cancels from the balance of a closed water network.

For a quadratic level curve, the formula requires no division:

$$\overline{Z}_i(a,b) = z_i + s_i \frac{a+b}{2} + c_i \left(\frac{(a+b)^2}{4} + \frac{(b-a)^2}{12} \right). \tag{7}$$

For a tabulated curve, we integrate its original linear segments, splitting only at crossed knots. An endpoint trapezoid would be wrong if an intervening knot changes the slope. The implementation also avoids subtracting nearby potentials when a and b almost coincide. Affine curves retain the usual midpoint head. With curvature, this head differs from $Z_i\big(\frac{a+b}{2}\big)$ by $c_i \frac{(b-a)^2}{12}$.

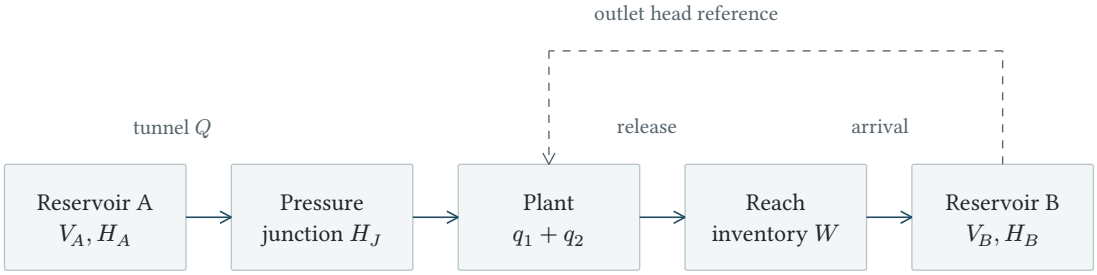


Figure 1: Solid arrows show where the water goes. The dashed connection specifies the plant's outlet head. Water reaches reservoir B after traveling through the river, but B's present level already affects generation.

3 Tunnel hydraulics and energy conversion

3.1 Tunnels and gates

For an oriented tunnel $e = (v, w)$, positive Q_e flows from v to w. With opening $b > 0$, the head difference equals friction loss:

$$H_{vt} - H_{wt} = K_e(b_{et}) Q_{et} |Q_{et}|,$$

$$K_e(b) = \begin{cases} \frac{R_e}{b} & \text{without a gate-loss curve} \\ R_e + G_e(b) & \text{with a supplied gate-loss curve.} \end{cases} \tag{8}$$

The tunnel resistance $R_e > 0$ and additional gate resistance $G_e \ge 0$ have units $\frac{\text{s}^2}{\text{m}^5}$. Positive Q requires $H_v > H_w$, and negative Q requires $H_v < H_w$. In either direction, friction dissipates energy with a loss proportional to Q^2. Flows through branches and loops follow from these head equations and the junction balances.

A closed gate permits a head difference while stopping flow:

$$b_{et} = 0 \Longrightarrow Q_{et} = 0, \quad |Q_{et}| \le C_{et}. \tag{9}$$

Equation (8) applies when the gate is open. Capacity C_{et} bounds flow in either direction. River outfalls that allow only discharge have $Q_{et} \ge 0$. A tunnel position may be prescribed or chosen from a finite

set $\mathcal{S}_e \subset [0,1]$. Prescribed positions may take any value in $[0,1]$, but the optimizer searches only the listed positions.

An instantaneous channel between reservoirs follows a supplied discharge law:

$$X_{ct} = \Phi_c\big(H_{vt} - H_{wt}, H_{\mathrm{ref},t}\big), \quad |X_{ct}| \le C_{ct}. \tag{10}$$

Φ_c is a family of curves indexed by a specified upstream or downstream reservoir level. Each curve gives discharge as a function of level difference, is nondecreasing and passes through $(0,0)$. Positive and negative responses need not be symmetric. We interpolate linearly along each curve and between reference levels. We extend the end segments beyond the tabulated level differences. Outside the reference-level range, we use the nearest curve. Water passes through the channel within the interval, without a separate transit inventory.

3.2 Energy conversion

Write $q_{pt} = \sum_{g \in \mathcal{G}_p} q_{gt}$ and $P_{pt} = \sum_{g \in \mathcal{G}_p} P_{gt}$ for plant totals. All units at a plant share the same effective head. Each plant has one operating mode, with $\sigma_p = 1$ for generation and $\sigma_p = -1$ for pumping:

$$h_{pt} = \sigma_p\Big(H_{\mathrm{source}(p),t} - \max\Big\{H_{\mathrm{target}(p),t}, h_p^{\mathrm{floor}}\Big\} - T_p\big(q_{pt}\big)\Big). \tag{11}$$

The target head is the outlet reference in Figure 1. If an outlet-level floor is specified, the maximum in (11) enforces it. Otherwise we use the target head alone. Total plant discharge raises the tailwater by $T_p\big(q_{pt}\big) \ge 0$. We model shared penstocks as tunnels, whose friction loss depends on the combined discharge of the units they supply. For pumping, (11) is destination head minus intake head plus the path loss.

Distinct penstocks can feed units that share a pressurized tailrace. Give each penstock its own intake node and connect the plant outlets to one junction. Conservation makes the tailrace tunnel carry their combined discharge. Its quadratic loss raises the common outlet head, reducing every unit's head once. These hydraulic supply groups remain separate plants for operating rules; a common outlet does not combine their power caps or commitment rules.

The binary state u_{gt} distinguishes a stopped unit from a committed unit:

$$u_{gt} \in \{0,1\}, \quad \underline{q}_{gt} u_{gt} \le q_{gt} \le \overline{q}_{gt} u_{gt}, \quad \underline{P}_{gt} u_{gt} \le P_{gt} \le \overline{P}_{gt} u_{gt}. \tag{12}$$

Lower flow and power limits may be zero. Commitment still records startup and dwell obligations when a committed unit produces no power.

The electrical output is the available hydraulic power multiplied by turbine and electrical efficiencies:

$$P_{gt} = \gamma q_{gt} h_{pt} \eta_g\big(q_{gt}, h_{pt}\big) e_g\big(P_{gt}\big). \tag{13}$$

For a pump, the hydraulic work is supplied by consumed electrical power. With constant motor efficiency $e_g^{\mathrm{motor}} \in (0,1]$, the equation is

$$P_{gt} \eta_{g(q_{gt}, h_{pt})} e_g^{\mathrm{motor}} = \gamma q_{gt} h_{pt}. \tag{14}$$

Lower efficiency therefore increases consumption. The same on/off and operating-domain constraints apply, with positive pumping lift required. Hydraulic efficiency tables retain their declared interpolation. Electrical motor-efficiency curves are outside this formulation. For a fixed-speed pump, one supplied discharge and hydraulic efficiency at each reference head define piecewise-linear functions $Q_{g(h)}$ and $\eta_{g(h)}$. Its equations are

$$q_{gt} = u_{gt} Q_{g(h_{pt})}, \quad P_{gt} \eta_{g(h_{pt})} e_g^{\mathrm{motor}} = \gamma q_{gt} h_{pt}. \tag{15}$$

No variable-discharge efficiency surface is inferred. Head coverage is required unless linear end-secant extrapolation is explicitly supplied. A single reference head requires an explicit constant characteristic. Static and operative unit bounds intersect this point, and an incompatible bound makes operation infeasible. Independent integration retains commitment and solves characteristic flow jointly with storage and pressure, including on finer replay grids. Schedule validation checks the original flow against the characteristic without replacing it. Equal discharge bounds on an ordinary variable-discharge unit instead prescribe a fixed-flow operating point.

When $u_{gt} = 1$, the unit must also satisfy head and efficiency limits and, when supplied by a turbine table, head-dependent discharge limits:

$$\begin{gathered} \underline{h}_g \le h_{pt} \le \overline{h}_g, \quad \eta_g^{\min} \le \eta_g(q_{gt}, h_{pt}) \le 1, \\ q_g^{\mathrm{lo}}(h_{pt}) \le q_{gt} \le q_g^{\mathrm{hi}}(h_{pt}). \end{gathered} \tag{16}$$

When $u_{gt} = 0$, (12) gives $q = P = 0$ and the operating inequalities in (16) are inactive. Plant head remains tied to the hydraulic node heads when all units are off. Isolated pressure components may retain a common reference freedom, as discussed below.

Turbine efficiency is a supplied table or the analytic relationship

$$\eta_g(q,h) = \eta_g^0 - \alpha_g \left(\frac{q - q_g^*}{q_g^*} \right)^2 - \beta_g \left(\frac{h - h_g^*}{h_g^*} \right)^2. \tag{17}$$

Here q_g^* and h_g^* are reference operating conditions, η_g^0 is their turbine efficiency. Nonnegative α_g, β_g describe a decrease away from them; signed coefficients are also admitted, subject to the same operating efficiency bounds. At reference head h_j, a supplied efficiency curve $f_{j(q)}$ has its own discharge knots. These grids may differ between heads. Interpolate each original curve linearly or with its original shape-preserving cubic slopes, then interpolate its value at the same discharge between heads:

$$\eta_{g(q,h)} = (1-b) f_{j(q)} + b f_{j+1}(q), \quad b = \frac{h - h_j}{h_{j+1} - h_j}. \tag{18}$$

A strict lookup between heads requires both contributing curves to cover q. At an exact reference head, only that curve contributes. For complete-case strict coverage, every column participating over the declared operating head range must cover the unit's static discharge range. When discharge continuation is explicitly allowed, each curve uses its own outer secant before head interpolation. Efficiency coverage does not define or modify the operating discharge envelope. A single reference head requires an explicit assumption that efficiency and the discharge envelope are independent of head. The actual hydraulic head still enters (13). Operating limits are supplied separately from efficiency knots, with explicit linear continuation required outside the data.

Turbine efficiency converts hydraulic power to shaft power. Electrical efficiency e_g accounts for the remaining conversion to electrical output. It is a separate piecewise-linear function of *electrical output* P, so (13) must be solved for P even when q, h are known. We require $P/e_g(P)$ to be strictly increasing over the admissible output range. The scalar power equation then has at most one admissible solution. If no electrical-efficiency curve is given, $e_g = 1$.

Plant state U_{pt} is one whenever at least one unit runs:

$$U_{pt} \ge u_{gt} \quad (g \in \mathcal{G}_p), \quad U_{pt} \le \sum_{g \in \mathcal{G}_p} u_{gt}, \quad U_{pt} \in \{0,1\}. \tag{19}$$

Plant limits apply to aggregate output and discharge: $P_{pt} \le \overline{P}_{pt}$ and $q_{pt} \le \overline{q}_{pt}$ when supplied, and $P_{pt} \ge \underline{P}_{pt} U_{pt}$, $q_{pt} \ge \underline{q}_{pt} U_{pt}$.

A single reversible machine has generating and pumping mode records on oppositely connected plants. An exclusion consists of supplied groups of unit names. For any two different groups K_a, K_b, we require

$$u_{it} + u_{jt} \le 1 \quad (i \in K_a, j \in K_b, a \ne b). \tag{20}$$

Parallel units within one group may run together. Initially running states must satisfy the same exclusion. Existing unit dwell histories, power and discharge histories carry into a restarted horizon without a separate pump state. Independently connected pumps and turbines may run simultaneously when no exclusion is supplied; shared ternary-machine characteristics are not modeled.

4 Outlets and river transport

4.1 Entry laws, confluences and splits

A river *reach* is a link with a defined entry and downstream destination. For a reach fed by a reservoir outlet, total entry is $R_{rt} = n_{rt} + d_{rt}$: natural reach inflow n plus reservoir withdrawal d. With outlet opening a_{rt}, one chosen law determines withdrawal:

$$d_{rt} = \begin{cases} (C_{rt} - n_{rt})a_{rt} & \text{controlled outlet} \\ c_r a_{rt}\sqrt{\ell_{rt}} & \text{orifice} \\ c_r \ell_{rt}^{\frac{3}{2}} & \text{uncontrolled weir} \\ a_{rt} D_r\left(H_{\text{source}(r),t}\right) & \text{supplied discharge curve.} \end{cases} \tag{21}$$

ℓ is the head above the outlet crest, $H_{\text{source}} - z_r^{\text{crest}}$. It must be nonnegative unless dry operation is allowed, in which case we use $[H_{\text{source}} - z_r^{\text{crest}}]_+$, with $[x]_+ = \max(x, 0)$. The coefficient c_r or curve D_r is supplied. An uncontrolled weir has $a = 1$, also when its law is given as a table. Other openings satisfy their bounds in $[0, 1]$. Ordinary reaches obey $n_{rt} \le R_{rt} \le C_{rt}$. The reservoir loses $d = R - n$. Natural reach inflow enters the river directly and must not be subtracted from reservoir storage.

The square-root derivative is singular at a dry outlet. Local dispatch instead uses $w \ge 0$ with $w^2 = [\ell]_+$. This piecewise polynomial equality gives exactly $w = \sqrt{[\ell]_+}$, so withdrawal is caw or cw^3. The global model uses a binary positive-part graph followed by $w^2 = [\ell]_+$. Both formulations retain the crest boundary without smoothing the physical law.

The analytic outlet laws are one-level responses; they do not calculate downstream backwater or gate geometry. A supplied one-level curve is valid only where the omitted conditions are fixed or accounted for in its calibration; it cannot describe independently varying downstream counterpressure. Directed river connections must be physically applicable over the operating range. Transport conservation does not enforce downhill flow or a gravitational energy balance along a routed reach. A plant or tunnel's hydraulic outlet reference must also represent its actual outfall head. Tunnel connections assume submerged entrances and exits.

At a zero-storage river node, incoming water is distributed among outgoing reaches with prescribed shares α_r. Local natural inflow and plant or tunnel outfalls enter each branch after this allocation. Let L_{rt} collect these local injections:

$$L_{rt} = n_{rt} + \sum_{g \to r} q_{gt} + \sum_{e \to r} Q_{et}, \quad R_{rt} = L_{rt} + \alpha_r \sum_{s \to r} \frac{A_{st}}{\kappa \Delta_t}. \tag{22}$$

Here $g \to r$ and $e \to r$ identify local outfalls, while $s \to r$ identifies reaches arriving at the source node of r. At every zero-storage node, $0 \le \alpha_r \le 1$ and the outgoing shares sum to one. These supplied constants hold throughout the horizon. A confluence with one outgoing reach has $\alpha_r = 1$. A 40% branch receiving 5 m³/s upstream and 1 m³/s locally carries 3 m³/s. Its capacity must accommodate both sources; spare capacity on another branch cannot change the prescribed split. Delayed reaches form a directed acyclic network, so we propagate arrivals in upstream order. Tunnels and instantaneous channels may form loops.

4.2 Release cohorts and travel-time responses

A *cohort* is the water entering a reach during one release interval. With a fixed travel time, that release simply appears downstream later. With a distributed delay, different fractions of the cohort arrive at different times. The response functions are deterministic approximations of river transport [3].

For reference entry flow ρ_{rj}, let fraction $w_{rj\ell}$ have a travel time uniformly distributed over $\left[\delta_{rj\ell}, \delta_{rj,\ell+1}\right]$. Delay edges must be nonnegative and strictly increasing. The fractions obey $w_{rj\ell} \ge 0$ and $\sum_\ell w_{rj\ell} = 1$. The cumulative fraction that has arrived after elapsed time s is

$$F_{rj}(s) = \sum_\ell w_{rj\ell} \min\left\{1, \max\left\{0, \frac{s - \delta_{rj\ell}}{\delta_{rj,\ell+1} - \delta_{rj\ell}}\right\}\right\}. \tag{23}$$

Entry flow may change the travel response. For two or more nonnegative, strictly increasing reference flows covering zero through reach capacity, interpolate between the neighboring distributions. A

missing zero-flow reference is rejected. Zero flow releases no water. In the following equations, SOS2 is a *special ordered set of type 2*: at most two weights may be nonzero, and their indices must be adjacent.

$$\theta_{rkj} \geq 0, \quad \sum_j \theta_{rkj} = 1, \quad R_{rk} = \sum_j \rho_{rj}\theta_{rkj}, \quad \theta_{rk} \in \mathrm{SOS2}\,. \tag{24}$$

The weights reproduce the cohort's total entry flow, including natural inflow and upstream arrivals. For example, halfway between two reference flows, the two distributions receive equal weights. A single distribution needs only $\theta_{rk1} = 1$. We then omit the flow-reproduction equation, so the reference flow does not fix the release. Once released, each cohort keeps its chosen mixture. Reference flows are calibration data and are unaffected by changes in operating capacity.

Consider water entering at time s in interval k. The fraction of it arriving during interval t is the cumulative fraction at age $\tau_{t+1} - s$ minus that at age $\tau_t - s$. Averaging over the release interval gives

$$B_{rtkj} = \frac{1}{\Delta_k} \int_{\tau_k}^{\tau_{k+1}} \left[F_{rj}(\tau_{t+1} - s) - F_{rj}(\tau_t - s)\right] \mathrm{d}s. \tag{25}$$

Multiplying by released volume and summing over cohorts gives arrival volume A_{rt}:

$$A_{rt} = A_{rt}^{\mathrm{hist}} + \kappa \sum_k \Delta_k R_{rk} \sum_j \theta_{rkj} B_{rtkj}. \tag{26}$$

Water arriving at clock time s from cohort k must have a travel time between $s - \tau_{k+1}$ and $s - \tau_k$. Integrating the delay density over this range gives the difference of cumulative fractions below. Multiplying by the cohort's release rate gives its contribution to instantaneous arrival rate:

$$f_r(s) = f_r^{\mathrm{hist}}(s) + \sum_k R_{rk} \sum_j \theta_{rkj} \left[F_{rj}(s - \tau_k) - F_{rj}(s - \tau_{k+1})\right]. \tag{27}$$

The historical terms account for releases before τ_0. We evaluate the integrals exactly for the uniform bins. Since θ depends linearly on R within a reference-flow interval, multiplying the arrival fraction by released volume gives a quadratic dependence on R. Interpolating arrival volumes directly would give a different travel response.

Each cohort selects θ from its own contemporaneous total release, including historical cohorts at their original flows. The nonlinear dispatch, global model and physical reconstruction use this same law. A frozen-reference MILP proposes commitments but provides no bound for this nonlinear problem. Physical releases stay within the supplied reference domain; endpoint holding in Newton trials and outer-polynomial extensions in nonlinear operators are numerical continuations only.

When all reaches have fixed travel times $\delta_r \geq 0$, we propagate the piecewise-constant entry stream $\tilde{R}_r(s)$ through the network. Each reach shifts its stream by its delay. At a river node, each outgoing stream receives its share of the shifted upstream streams:

$$\begin{gathered} f_r(s) = \tilde{R}_r(s - \delta_r), \quad \tilde{R}_r(s) = L_r(s) + \alpha_r \sum_{s' \to r} f_{s'}(s), \\ A_{rt} = \kappa \int_{\tau_t}^{\tau_{t+1}} f_r(s)\,\mathrm{d}s. \end{gathered} \tag{28}$$

The allocation relation applies for $s \geq 0$. At a reservoir source, $\tilde{R}_r(s) = n_{rt} + d_{rt}$ on $[\tau_t, \tau_{t+1})$. For $s < 0$, it is the independently supplied entry history of that reach, already allocated to its branch. Upstream historical water arriving at a node after zero is allocated once, on entry to the outgoing reaches. Interval entry variables are the averages of these streams:

$$R_{rt} = \frac{1}{\Delta_t} \int_{\tau_t}^{\tau_{t+1}} \tilde{R}_r(s)\,\mathrm{d}s. \tag{29}$$

For example, a one-hour release passing through two reaches with half-hour delays must arrive as a one-hour release shifted by one hour. On an hourly mixing grid, averaging after the first reach would spread some arrival before and after that final interval. A release can reach downstream within its control interval when the delay is zero or shorter than that interval. We check entry capacities on the stream, including both sides of each jump.

If any reach has a distributed delay, (22) instead uses the mean upstream arrivals as the next reach's interval release. Equations (25) and (26) are exact for those rectangular cohorts, but this downstream averaging approximates the continuous timing. We assess this error on a finer transport grid by comparing cumulative arrivals and water in transit. A volume balance cannot detect water that arrives at the wrong time.

4.3 Water in transit, history and environmental flows

Water released but not yet delivered belongs to reach inventory W. Its change equals entry volume minus arrival volume:

$$W_{r,t+1} = W_{rt} + \kappa \Delta_t R_{rt} - A_{rt}, \quad W_{r0} = W_r^{\text{hist}}. \tag{30}$$

Pre-horizon entry records determine both initial inventory and future historical arrivals. For a past cohort k released on $[s_k, e_k)$, let F_k be its selected delay mixture. Water released at $s < 0$ has age $-s$ at the horizon start, so the fraction still in transit is $1 - F_k(-s)$:

$$W_r^{\text{hist}} = \kappa \sum_{k \in \mathcal{H}_r} R_{rk} \int_{s_k}^{e_k} [1 - F_k(-s)] \, \mathrm{d}s. \tag{31}$$

The historical cohorts of reach r form $\mathcal{H}_r$. We assume zero entry before the supplied history. A reach's records already include past upstream contributions, so propagating them again would count the water twice. To restart after executing part of a schedule, we save its entry cohorts and flows as well as reservoir states and operating histories. Average arrivals alone cannot determine when the water still in transit will arrive.

Summing reservoir and reach balances cancels every internal transfer:

$$\begin{aligned} &\sum_i (V_{i,t+1} - V_{it}) + \sum_r (W_{r,t+1} - W_{rt}) \\ &\quad = \kappa \Delta_t \left(\sum_i I_{it} + \sum_r n_{rt} + E_t \right). \end{aligned} \tag{32}$$

Here E_t is net boundary exchange, positive into the modeled system. For a plant outfall, the source reservoir loses q and the reach receives it. Arrival A later leaves the reach and enters the receiving reservoir. Both transfers cancel in the sum. Total stored and in-transit water changes only through natural inflows and external exchanges, including when an instantaneous connection carries reversed flow.

Environmental requirements can constrain reach entry $R_{rt} \geq R_{rt}^{\min}$, average arrival over a specified time window J, or arrival at every instant:

$$\int_J f_r(s) \, \mathrm{d}s \geq \int_J f_r^{\min}(s) \, \mathrm{d}s, \quad f_r(s) \geq f_r^{\min}(s). \tag{33}$$

The supplied policy chooses the window or pointwise arrival constraint. Distributed arrival rates from rectangular cohorts are piecewise linear. Fixed-delay streams are piecewise constant. Between combined response and requirement breakpoints, the difference $f_r - f_r^{\min}$ is affine, so its minimum occurs at an endpoint. It is enough to check these endpoints, taking both one-sided limits at a jump. At the horizon endpoints, use the limit from inside the horizon.

An aggregate flow requirement can also combine named unit discharges, river entries, channel flows and a given inflow contribution I_{ot}:

$$\sum_{g \in \mathcal{G}_o} q_{gt} + \sum_{r \in \mathcal{D}_o \cup \mathcal{C}_o} R_{rt} + I_{ot} \geq M_{ot}. \tag{34}$$

The sets $\mathcal{G}_o$, $\mathcal{D}_o$ and $\mathcal{C}_o$ select the contributions, and M_{ot} is the required minimum. Channel flow $R_{ct} = X_{ct}$ is signed. The inequality constrains existing flows and adds no water to the balances.

5 Operating constraints over time

A unit's initial operating state and its duration in that state are given. Binary starts s^+ and stops s^- record changes between consecutive intervals:

$$u_{gt} - u_{g,t-1} = s^+_{gt} - s^-_{gt}, \quad s^+_{gt} + s^-_{gt} \le 1, \quad s^\pm_{gt} \in \{0,1\}. \tag{35}$$

Minimum running time L^{up}_g means that a start at τ_t requires $u_{gk} \ge s^+_{gt}$ for each later interval with $0 \le \tau_k - \tau_t < L^{\text{up}}_g$. Minimum stopped time L^{down}_g similarly requires $u_{gk} \le 1 - s^-_{gt}$ whenever $0 \le \tau_k - \tau_t < L^{\text{down}}_g$. These tests use elapsed hours, including on an irregular grid. Initial state age determines any remaining mandatory running or stopped time.

Apply these minimum-duration constraints to U if they are also specified for the plant. A handover between running units does not restart the plant. Any minimum duration unfinished at the horizon end becomes a requirement on the next schedule.

Availability, forced states and prescribed output restrict the same decisions:

$$\underline{u}_{gt} \le u_{gt} \le \overline{u}_{gt}, \quad P_{gt} = P^{\text{given}}_{gt}, \quad q_{gt} = q^{\text{given}}_{gt}. \tag{36}$$

We impose an output schedule only where one is supplied. Plant schedules constrain the sums of unit outputs, and plant state obeys $\underline{U}_{pt} \le U_{pt} \le \overline{U}_{pt}$. Maintenance requires units to be off. Zero scheduled output requires the off state, and positive output requires the on state. These constraints also apply during mandatory initial running or stopped periods. A conflict makes the model infeasible.

For interval-average discharge or release x_t, supplied upward and downward rates r^+_x, r^-_x limit changes between interval midpoints:

$$-\int_{m_{t-1}}^{m_t} r^-_x(s)\,\mathrm{d}s \le x_t - x_{t-1} \le \int_{m_{t-1}}^{m_t} r^+_x(s)\,\mathrm{d}s. \tag{37}$$

A running unit cannot enter its positive operating range from zero when the allowed inter-interval increase is smaller than its minimum output. We therefore allow a minimum-output jump at startup and shutdown in the power-ramp bounds. Add $\sum_{g\in\mathcal{G}_x} \underline{P}_{gt} s^+_{gt}$ to the upward allowance and $\sum_{g\in\mathcal{G}_x} \underline{P}_{g,t-1} s^-_{gt}$ to the downward allowance. Here $\mathcal{G}_x$ contains that unit or all units at the constrained plant. Discharge and river-release ramps have no jump allowance. For the first interval, use the average and midpoint of the supplied previous window. Omit that comparison if no previous window is supplied.

Reservoir volume and level ramps instead constrain vertex changes $V_{i,t+1} - V_{it}$ and $Z_i(V_{i,t+1}) - Z_i(V_{it})$, with rate integrals over $[\tau_t, \tau_{t+1}]$. The ramp limits apply between recorded states.

A tunnel interlock limits how many gates in a group $\mathcal{E}_\ell$ may be open at once. With $o_{et} = \mathbf{1}[b_{et} > 0]$,

$$\underline{N}_{\ell t} \le \sum_{e\in\mathcal{E}_\ell} o_{et} \le \overline{N}_{\ell t}. \tag{38}$$

A positive position counts as open even when its flow is zero.

Some handovers also require a plant shutdown after a gate closes. For each interlock, specify protected plants $\mathcal{P}_\ell$ and disjoint closure groups $\mathcal{E}^d_\ell \subseteq \mathcal{E}_\ell$, with d denoting forward or backward. Each group has a supplied waiting time $L^d_\ell \ge 0$ in hours. Closing its gate at τ_t requires every protected unit to be off during the wait:

$$\begin{gathered} u_{gk} + o_{e,t-1} - o_{et} \le 1, \quad \tau_t \le \tau_k < \tau_t + L^d_\ell, \\ e \in \mathcal{E}^d_\ell, \quad g \in \cup_{p\in\mathcal{P}_\ell} \mathcal{G}_p. \end{gathered} \tag{39}$$

At closure, the gate-status difference is one, so the inequality forces $u_{gk} = 0$. For every other transition it adds no restriction. It uses the existing variables, with initial gate status supplied before the first interval. Another intake may open immediately while the plant remains idle. Movement between positive positions is not a closure, and reopening does not cancel a wait. Controls change only at grid edges; a wait expiring inside an interval permits generation at the next edge.

A pre-horizon closure can leave a residual wait $w^0_{\ell d}$:

$$0 \le w^0_{\ell d} \le L^d_\ell, \quad u_{gk} = 0 \quad \text{when} \quad \tau_k - \tau_0 < w^0_{\ell d}. \tag{40}$$

For a positive residual, the protected units must initially be off with off-age at least $L^d_\ell - w^0_{\ell d}$. This makes their histories consistent with the implied closure time. Zero declares no pending wait. On restart we retain the latest unexpired closure deadline in each group, together with the gate positions and unit histories. Closing a gate at the restart edge starts a new wait in the continuation.

5.1 Explicitly penalized operating limits

A separate named rule may permit bounded deviations from a reservoir observation or a unit or plant output target. Existing operating schedules remain hard unless replaced explicitly by such a rule. For an observed quantity x_ℓ, supplied limits L_ℓ, U_ℓ use nonnegative slacks:

$$x_\ell + \sigma_\ell^- \geq L_\ell, \quad x_\ell - \sigma_\ell^+ \leq U_\ell, \quad 0 \leq \sigma_\ell^\pm \leq \overline{\sigma}_\ell^\pm. \tag{41}$$

Omit a side when its limit is absent. A zero allowance makes that side hard. Positive allowances have strictly positive penalty coefficients. A soft output target adds no commitment implication; generation can remain off and incur the permitted shortfall.

Reservoir volume observations use $x_\ell = V_i(\theta_\ell)$ and level observations use $x_\ell = Z_i(V_i(\theta_\ell))$ at a declared absolute grid vertex θ_ℓ. Their slacks have units Mm^3 and metres. A terminal target is an observation at the original terminal time, which is retained when the case is continued. These rules constrain the recorded states and make no assertion about a continuous-time band between observations.

For power and discharge, observe the average over an original window $[a_\ell, b_\ell]$ whose endpoints are grid edges:

$$x_\ell = \frac{1}{b_\ell - a_\ell} \sum_{t : a_\ell \leq \tau_t < \tau_{t+1} \leq b_\ell} \Delta_t y_t. \tag{42}$$

Here y_t is one unit's output or the sum of the plant's member outputs. The slacks have units MW or m^3/s. Separate windows are required when one part of a window must not offset another.

The nonnegative cost is

$$C_\ell = \alpha_\ell (c_\ell^- \sigma_\ell^- + c_\ell^+ \sigma_\ell^+), \quad \alpha_\ell = \begin{cases} 1 & \text{reservoir volume or level observation} \\ b_\ell - a_\ell & \text{power window} \\ \kappa(b_\ell - a_\ell) & \text{discharge window} \end{cases}. \tag{43}$$

Costs are therefore supplied per Mm^3, per metre, per MWh or per Mm^3, respectively. State observation costs are charged once at the observation, without a duration multiplier. Finer replay keeps these same observations and windows and does not create additional penalty terms. Restart drops completed assessments, retains future times and rejects a restart inside an output window because partial assessment history is not part of this input profile.

Water balance and every constitutive relation remain hard, together with the static domains needed for valid equipment calibration and global bounds. A reservoir volume rule can tighten its domain only to the expanded limits $L_\ell - \overline{\sigma}_\ell^-$ and $U_\ell + \overline{\sigma}_\ell^+$, intersected with its hard safety domain. The nominal limits alone cannot enclose all admitted solutions. Independent assessment reports the actual minimal slacks $\max(0, L_\ell - x_\ell)$ and $\max(0, x_\ell - U_\ell)$ and their costs. A permitted positive slack is compatible with an accepted schedule; exceeding an allowance or violating the physics is not.

All permitted deviations and their costs are explicit case data. Zero allowances retain hard limits; no implicit endpoint penalty is added.

6 Spinning reserve capacity

A reserve group $k \in \mathcal{K}$ has eligible generating units $\mathcal{G}_k$, upward and downward requirements $D_{kt}^\pm$, capacity ceilings $\overline{R}_{kt}^\pm$ and response time $\theta_k > 0$ in hours. Allocation $r_{gkt}^\pm$ is MW. It is zero outside $\mathcal{G}_k$ and obeys reserve availability $a_{gt}^{\mathrm{res}} \in \{0, 1\}$:

$$0 \leq r_{gkt}^\pm \leq a_{gt}^{\mathrm{res}} \overline{P}_g u_{gt}, \quad \sum_g r_{gkt}^\pm \leq \overline{R}_{kt}^\pm. \tag{44}$$

Here $\overline{P}_g$ is nameplate power. Omitted group ceilings use the sum of eligible nameplate power limits. A group requirement is hard unless a positive shortfall cost is explicitly supplied:

$$\sum_g r_{gkt}^\pm + \sigma_{kt}^\pm \geq D_{kt}^\pm, \quad 0 \leq \sigma_{kt}^\pm \leq D_{kt}^\pm. \tag{45}$$

For a hard requirement $\sigma = 0$. Positive shortfall costs make it equal $\left[D_{kt}^\pm - \sum_g r_{gkt}^\pm\right]_+$ at optimum. Independent assessment recomputes that deficit from allocations and checks the stored value.

The following reserve operating-point equations apply to generating stations. Pumps have zero reserve allocation and retain their nominal flow and consumption, $q^{\pm}_{gt} = q_{gt}$ and $P^{\pm}_{gt} = P_{gt}$. All groups share each generating unit's physical headroom:

$$\begin{gathered} R^{\pm}_{gt} = \sum_{k} r^{\pm}_{gkt}, \quad P^{+}_{gt} = P_{gt} + R^{+}_{gt}, \quad P^{-}_{gt} = P_{gt} - R^{-}_{gt}, \\ q^{+}_{gt} \geq q_{gt}, \quad q^{-}_{gt} \leq q_{gt}. \end{gathered} \tag{46}$$

Both operating points retain the nominal commitment u. They satisfy (12), (13) and (16) with their respective q^{+}, P^{+}, h^{+} or q^{-}, P^{-}, h^{-}. The plant totals obey the same operative power/discharge bounds and plant state U as nominal generation. Downward capacity therefore cannot stop a unit or cross its minimum operating output. No monotonic relationship between power and discharge is assumed.

The two points hold the scheduled hydraulic node heads fixed, while shared station tailwater is re-evaluated:

$$h^{\pm}_{pt} = H_{\mathrm{source}(p),t} - \max\left\{H_{\mathrm{target}(p),t}, h^{\mathrm{floor}}_{p}\right\} - T_{p}\left(\sum_{g \in \mathcal{G}_{p}} q^{\pm}_{gt}\right). \tag{47}$$

Every operating unit in the station participates in these power equations. A unit with no allocation may change its witness flow to hold nominal power when another unit changes shared tailwater. Summing allocations in (46) prevents selling the same headroom in multiple groups.

Let θ_{p} be the minimum θ_{k} among groups with at least one eligible unit in the station. For a unit or plant output $x \in \{P, q\}$, response rates use the same supplied directional rates as (37), with no startup/shutdown jump:

$$x^{+}_{t} - x_{t} \leq \int_{m_{t}}^{m_{t}+\theta_{p}} r^{+}_{x}(s)\,\mathrm{d}s, \quad x_{t} - x^{-}_{t} \leq \int_{m_{t}}^{m_{t}+\theta_{p}} r^{-}_{x}(s)\,\mathrm{d}s. \tag{48}$$

The station-wide response is conservative for products with different response times and also limits units compensating another group's tailwater. Finer replay retains these physical response limits at its operating points.

Optional sustained activation duration $\delta^{\mathrm{act}}_{k} \geq 0$ is distinct from response time. A positive duration requires supplied unit qualifications E^{+}_{g}, E^{-}_{g} in MWh:

$$\sum_{k} \delta^{\mathrm{act}}_{k} r^{+}_{gkt} \leq E^{+}_{g}, \quad \sum_{k} \delta^{\mathrm{act}}_{k} r^{-}_{gkt} \leq E^{-}_{g}. \tag{49}$$

These qualifications apply independently to each interval. They are not a consumable reservoir inventory and remain unchanged at restart. Capacity operating points add no flow or energy activation to nominal water balances. They do not re-solve tunnel losses, storage evolution, river transport or environmental flows for an activated trajectory. Cascade activation requires a separate treatment, such as the two-stage stochastic and robust models studied by Naversen et al. [4]. A certificate here concerns only the declared capacity model at scheduled node heads. Pumping reserves, nonspinning starts/stops, frequency-response droop, forbidden-zone transitions and reserve market purchases are outside its scope.

7 Economic objective

Generation at price π_t per MWh earns $\pi_t \Delta_t P_{gt}$; pumping consumption incurs that amount. Let σ_g be the sign of the unit's plant mode. Net electrical exchange is $\sum_g \sigma_g P_{gt}$. Let c^+, c^- be unit transition costs, c^b nonnegative gate-movement costs, c^r nonnegative release-shortfall costs, and v_i, v_r the values assigned to water in reservoirs and reaches, per Mm^3. We maximize

$$\begin{aligned}\mathcal{V} = &\sum_t \pi_t \Delta_t \sum_g \sigma_g P_{gt} - \sum_{g,t} \left(c^+_{gt} s^+_{gt} + c^-_{gt} s^-_{gt}\right) \\ &- \sum_{e,t} c^b_{et} \left| b_{et} - b_{e,t-1} \right| - \sum_{r,t} c^r_{rt} \xi_{rt} - \sum_\ell C_\ell \\ &+ \sum_i v_i (V_{iT} - V_{i0}) + \sum_r v_r (W_{rT} - W_{r0}) \\ &+ \sum_{k,t} \Delta_t \left(\pi^+_{kt} \sum_g r^+_{gkt} + \pi^-_{kt} \sum_g r^-_{gkt} - c^{\mathrm{res}}_k (\sigma^+_{kt} + \sigma^-_{kt}) \right).\end{aligned} \tag{50}$$

Starts, stops and gate movements are charged once per event, without a duration multiplier. The first gate comparison uses its supplied initial position. The reservoir-storage and reach-inventory terms value changes in stored water. A late release can still have value downstream after the horizon ends. The terminal values v_i, v_r are inputs that account for this later use. Reserve prices $\pi^{\pm}_{kt}$ and positive optional shortfall costs c^{res}_k are currency per MW-hour. Allocated MW earns capacity price times interval hours, without implying actual activation energy. Hard reserve requirements have zero shortfall and no penalty term.

An energy value e_i in currency per MWh can be converted at a supplied reference net head and combined efficiency:

$$f_i = 2.725 h^{\mathrm{ref}}_i \eta^{\mathrm{ref}}_i, \quad v_i = e_i f_i + v^{\mathrm{downstream}}_i. \tag{51}$$

f_i is MWh per Mm^3. The downstream term includes only opportunities not already represented by f_i. Water that has passed one station retains only its remaining opportunities. Heads, efficiencies, paths and exchange rates are not inferred. All monetary inputs use one currency. These constants describe future use; they are not marginal values calculated by optimization. Historical transit is included in W_{r0}, and subtracting initial inventory values preserves additive accounting across restart.

A positive release penalty permits a deficit below the entry minimum. The deficit variable is measured as a volume:

$$\xi_{rt} \geq 0, \quad \xi_{rt} \geq \kappa \Delta_t (R^{\min}_{rt} - R_{rt}). \tag{52}$$

A positive cost makes ξ equal the actual shortfall at optimum. Without that penalty, the entry minimum is hard and $\xi = 0$. Arrival minima and aggregate flow requirements remain hard. Subtracting the values of the initial inventories removes a constant from the reported objective and leaves the optimal decisions unchanged.

8 Solving the model and checking a schedule

8.1 Finding feasible controls

We build the models in JuMP [5]. SCIP first solves a MILP approximation to propose unit states and gate positions. We fix those choices and use Ipopt [6] to solve for the continuous decisions under the nonlinear hydraulic laws. The procedure may try several proposals within a time budget. Candidates that pass the checks below become starting solutions for the full SCIP search [7], which can change the discrete choices.

The MILP objective approximates (50) and gives no enclosing bound. Its soft level observations use a secant approximation to storage-dependent level, and its power observations use the proposal's power surrogate. The nonlinear dispatch and global model retain the original observation laws. Dispatch with the discrete choices fixed is a piecewise-smooth, nonconvex problem. A stationary solution need not be globally optimal.

8.2 Reconstructing the water trajectory

An independent integration holds the selected unit discharges and openings fixed and recovers the resulting water trajectory. At each interval, Newton's method solves jointly for ending storage, junction heads, tunnel flows and active outlet releases. Column-pivoted QR handles rank deficiency in the linearized equations, including cycles with zero tunnel flow. With a delay shorter than the interval, some current release arrives before the interval ends and belongs in the same hydraulic solve. Equation (13) then gives electrical output.

In some isolated pressure components, the equations determine only head differences. A common shift in all heads leaves the flows unchanged. We choose one reference head for the numerical solve and keep every water balance. We can then shift the recovered heads to satisfy the junction bounds if the bounds admit such a shift. Replay retains the candidate's pressure reference where the equations permit this freedom. An idle committed unit can still have an operating head bound relative to another component, so choosing a different reference can wrongly make its operating point appear inadmissible. The reference does not replace any hydraulic equation or absolute head restriction.

The nonlinear step need not have a unique solution. For example, a channel can be passive at each reference level while its conductance decreases so strongly with that level that the interval equations have several roots. Newton convergence alone does not select the continuous physical trajectory. Finer time steps are needed to assess such cases; a failed replay does not establish mathematical infeasibility.

Even a constant channel conductance K can cause coarse-grid overshoot. With equal affine level slopes s, the difference d in reservoir levels obeys $d_{\mathrm{new}} = \frac{1-\theta}{1+\theta} d_{\mathrm{old}}$, where $\theta = \kappa \Delta K s$. For $\theta > 1$, the levels cross during a discrete step; continuous passive equalization approaches equilibrium without crossing.

We accept a reconstructed schedule only if it satisfies the interval equations and operating restrictions. We repeat the integration on a finer grid using the same controls. Schedules and ramps are checked over their original operating windows. We compare storage at the original vertices, total water, cumulative arrivals and terminal inventory. For the specified volume tolerance ε_V and relative objective tolerance $\varepsilon_{\mathcal{V}}$, the reconstruction must satisfy

$$\begin{gathered} \max_{i,t} \left| V_i^{\mathrm{fine}}(\tau_t) - V_{it} \right| \le \varepsilon_V, \quad \max_{r} \left| W_r^{\mathrm{fine}}(\tau_T) - W_{rT} \right| \le \varepsilon_V, \\ \left| \mathcal{V}^{\mathrm{fine}} - \mathcal{V} \right| \le \varepsilon_{\mathcal{V}} \max(1, |\mathcal{V}|). \end{gathered} \tag{53}$$

The comparison estimates the effect of time discretization for the chosen controls. It does not give a rigorous bound on continuous-time hydraulic error.

The discrete-gradient chain rule removes artificial storage-cycle work. For direct internal transfers, multiply each node balance by its own interval head and sum: the internal edge terms become minus transferred volume times source-minus-target head. Passive tunnels and channels dissipate this work. Ideal pumping and generating units exchange it without loss; efficiencies below one add conversion losses. Summing potential changes around a closed storage cycle gives zero, even when the interval lengths differ.

This identity concerns storage and direct transfers. A delayed river carries no mechanical-energy state in this model. Its release and arrival energy ports do not cancel as an instantaneous transfer, and the storage chain rule does not establish an energy balance for the omitted river dynamics. Nor does energy consistency remove coarse-step overshoot or nonunique roots. The interval-model objective and its bound still concern an approximation to continuous physical time.

8.3 Bounding the remaining objective improvement

A feasible schedule gives an achievable objective value L. Global search provides an upper bound U that no feasible schedule in the declared model can exceed. SCIP obtains such bounds from relaxations, tightens them and branches on discrete choices or continuous variable ranges. A region can be discarded when its bound cannot improve the current feasible value.

A *relaxation* enlarges the feasible set. Its maximum is an upper bound on the scheduling objective because every feasible schedule is still in that set. The enlarged set may allow flow, head and efficiency values whose power exceeds the physical conversion law. A *valid inequality* holds for every feasible schedule and can exclude some of these excessive values.

The algebraic curve representations and bound constructions are given in Appendix A.

For numerical bounding, each scalar equality $f(x) = b$ is enclosed by $b - \delta \le f(x) \le b + \delta$, with $\delta = 10^{-9}$ in the equation's model coordinates. The solver feasibility tolerance is 10^{-10}. This explicit band prevents small inconsistencies between floating-point curve graphs from making the bounding problem empty. It enlarges the feasible set, so its upper bound also bounds exact equality solutions. The band is used only in global search; reconstruction and acceptance use the original equations. It does not enclose every error in coefficient arithmetic or in the solver.

Here L is the independently reconstructed value of a schedule feasible for the interval model, and U must be a finite enclosing bound. The absolute and relative limits on possible improvement are

$$g_{\text{abs}} = \max(0, U - L), \quad g_{\text{rel}} = \frac{\max(0, U - L)}{\max(1, |L|)}. \tag{54}$$

If $L = 100$ and $U = 103$, improvement is limited to 3 monetary units, or 3% under this definition. There may be no better schedule at all. The constant subtraction of initial water values in (50) affects the relative gap's denominator, so L and U must use that same objective.

A certificate states that the requested absolute or relative gap has been met for the interval model, interpolation rules and finite gate sets. Fixing discrete choices makes the certificate conditional on those choices. A schedule may meet the interval-model gap and fail the finer-grid checks. Neither test certifies a continuous-time optimum. The numerical certificate uses solver tolerances and floating-point bounds. It does not rigorously enclose every rounding error in the solver and polynomial calculations.

9 Discussion

The hydraulic equations let changes in reservoir levels and shared tunnel flows affect dispatch during optimization. River inventory accounts for released water until it reaches the next node. Using these nonlinear laws can be computationally expensive. Even with commitment fixed, the continuous problem is nonconvex, and a large network may take substantial time to close the objective gap.

Model accuracy also depends on the input curves. Finer integration can expose an inadequate time grid, but it cannot correct a miscalibrated efficiency or travel-time response. Tunnel positions absent from the declared finite set are excluded from the search. The objective certificate applies to this finite model, with its specified curves and control grid. Changing those assumptions requires solving a new problem.

Appendix

A Representing curves and strengthening bounds

The scheduling problem is a mixed-integer nonlinear program (MINLP). Fixing unit states and gate positions still leaves a nonconvex problem in flow, head, efficiency and travel response. SCIP needs algebraic equations for these functions and relaxations from which to compute upper objective bounds.

A.1 Representing supplied curves exactly

For a curve $y = f(x)$, the *graph* is the set of pairs (x, y) satisfying that equation. We reproduce the specified interpolant in the optimization model. Exactness here refers to the interpolant, not to the watercourse it approximates. A piecewise-linear curve with knots (x_i, y_i) has the graph

$$\lambda_i \ge 0, \quad \sum_i \lambda_i = 1, \quad x = \sum_i x_i \lambda_i, \quad y = \sum_i y_i \lambda_i, \quad \lambda \in \text{SOS2}\,. \tag{55}$$

SOS2 restricts the weights to one segment [8]. The two neighboring weights sum to one, and the equation for x fixes the interpolation fraction. The equation for y then gives the interpolated output. Without adjacency, weights on distant knots could produce points away from the curve.

For a reservoir level table, integrate each linear cell to obtain a quadratic potential. The endpoint level and potential share the same coordinate weights. Writing $\delta v_k = v_{k+1} - v_k$ and $s_k = \frac{Z_{k+1} - Z_k}{\delta} v_k$, the potential graph is

$$\Psi(v) = \sum_k \Psi(v_k)\lambda_k - \sum_k \frac{s_k \delta v_k^2}{2}\lambda_k \lambda_{k+1}. \tag{56}$$

The correction removes the quadratic's chord error. For endpoint volumes a, b, impose the work equation and a bounded divided-difference row:

$$(b-a)H = \Psi(b) - \Psi(a), \quad H = Z(a) + D(b-a), \quad \frac{\min_k s_k}{2} \le D \le \frac{\max_k s_k}{2}. \tag{57}$$

The slope bounds range over cells intersecting the permitted storage domain. They follow from $D = \int_0^1 (1-\theta) Z'(a + \theta(b-a))\,\mathrm{d}\theta$. For $a \neq b$, the first row fixes the mean head. At $a = b$, it is only $0 = 0$; the second row fixes $H = Z(a)$ without a direction binary or a volume-change threshold. Local dispatch and replay evaluate the segment integral directly. The global model subtracts one common elevation reference from Z and integrates that shifted level, which leaves the equations unchanged while reducing cancellation. Polynomial storage uses (7) directly.

For a turbine table, let q_i be the sorted union of the original discharge knots. These coordinates partition the interpolants; they are not newly measured calibration points. Discharge weights λ_i and head weights μ_j each satisfy the corresponding SOS2 and coordinate constraints. Let $E_{ij} = f_{j(q_i)}$ be the original curve's value at a partition point. Linear interpolation in head combines the values η_j on the neighboring head curves. For head-independent data, each head vertex uses the same efficiency column. A shape-preserving cubic in discharge [9] is represented by adding corrections to the linear curve:

$$\begin{gathered} w_i = \lambda_i \lambda_{i+1}, \quad r_i = \lambda_i w_i, \\ \eta_j = \sum_i E_{ij}\lambda_i + \sum_i \big[A_{ij} r_i + B_{ij}(w_i - r_i)\big], \quad \eta = \sum_j \mu_j \eta_j. \end{gathered} \tag{58}$$

For segment width $\delta q_i = q_{i+1} - q_i$, secant slope $d_{ij} = \big(E_{i+1,j} - E_{ij}\big)/\delta q_i$ and original polynomial derivatives $m_{ij}^{\text{left}}, m_{ij}^{\text{right}}$ at the cell's two endpoints, use

$$A_{ij} = \delta q_i \big(m_{ij}^{\text{left}} - d_{ij}\big), \quad B_{ij} = \delta q_i \big(d_{ij} - m_{ij}^{\text{right}}\big). \tag{59}$$

On the active segment, the two correction terms are multiples of $\lambda_i^2 \lambda_{i+1}$ and $\lambda_i \lambda_{i+1}^2$. They vanish at the endpoints and adjust the derivatives to match the original slopes. The variables w_i, r_i express the corrections through quadratic equalities. Setting $A = B = 0$ gives bilinear interpolation. Each original cubic is re-expressed on its union-grid subcells, retaining the derivatives from that polynomial. Recalculating PCHIP slopes from the partition-point values would generally change the interpolant and is not performed. Endpoint derivatives are taken from within each selected cell, including the linear continuation outside its original discharge data. Restricting the curve's domain leaves its cubic slopes unchanged. We use the declared end secants for discharge outside the table and for head extrapolation where allowed.

An off unit can have a network head outside its running table. A finite extension lets us evaluate the graph there. Its running head and efficiency limits are inactive, and $q = P = 0$.

Units with the same net-head variable and compatible table grids can share head weights. Each release cohort uses the same flow weights for arrivals, environmental checks and terminal inventory. Channel responses use a piecewise-bilinear graph. The product equalities in (58), (13) and (26) remain nonlinear.

A.2 Quadratic equations for tunnels and gates

A binary direction variable separates positive and negative tunnel flow:

$$\begin{gathered} Q = Q^+ - Q^-, \quad 0 \le Q^+ \le Cz, \quad 0 \le Q^- \le C(1-z), \\ S = (Q^+)^2 - (Q^-)^2, \quad z \in \{0, 1\}. \end{gathered} \tag{60}$$

At $z = 1$, flow is nonnegative and $Q^- = 0$. At $z = 0$, flow is nonpositive and $Q^+ = 0$. This gives $S = Q|Q|$ in either case. Directed outfalls use $S = Q^2$. Known head ordering can also fix the direction.

For each finite gate position, a binary selector identifies whether that position is chosen. Exactly one selector is active, so b, the open state o and the selected resistance $K(b)$ are weighted sums of

known values. To impose the loss equation only when the gate is open, define $d_o = o(H_v - H_w)$. If $L \le H_v - H_w \le U$, this binary product is represented by

$$Lo \le d_o \le Uo, \quad H_v - H_w - U(1-o) \le d_o \le H_v - H_w - L(1-o). \tag{61}$$

When $o = 0$, (61) forces $d_o = 0$. When $o = 1$, it forces $d_o = H_v - H_w$. Adding $|Q| \le Co$ and $d_o = K(b)S$ gives the open and closed gate laws in (8) and (9). The resistance assigned to the closed selector may be any finite value, since $S = 0$ there. All the product equalities are quadratic.

The outlet law also has a binary wet/dry branch for $\ell = \max(H - z^{\text{crest}}, 0)$. Its square root in (21) can be written using $r \ge 0$ and $r^2 = \ell$.

A.3 Upper bounds for global optimization

We bound variables using storage and equipment limits. Water balances propagate initial and terminal restrictions through the horizon to narrow these ranges. The bounds must hold for every admissible schedule.

We omit unit and time indices in the following bounds. These strengthening rows apply only to generating modes. Pump rows retain the original bounded work equality (14), without applying generation supports to consumption. For a running generating unit with nonnegative $qh\eta$, let $\overline{e}$ bound its electrical efficiency. Then $p^{\text{ub}}(q,h) = \gamma\overline{e}qh\eta(q,h)$ is an upper bound on physical electrical power. It is polynomial within each interpolation cell even when (13) is implicit in P. Suppose a plane $aq + bh + c$ bounds this polynomial throughout the running domain, with $b \ge 0$. For a head lower bound h_* valid in both operating states, the following inequality includes the off state:

$$P \le aq + b(h - h_*) + (c + bh_*)u. \tag{62}$$

When the unit runs, it reduces to $P \le aq + bh + c$. When it is off, $P = q = 0$ and the right side is nonnegative. To establish c, we bound the residual $p^{\text{ub}} - aq - bh$ over every original interpolation cell. In a Bernstein representation of a polynomial on a box, the basis weights are nonnegative and sum to one, so its value cannot exceed its largest coefficient. The coefficient bound covers the whole cell. Fitting a plane at a few samples alone could miss a larger residual between them.

A box containing the admissible running values of q, h, η has up to eight corners (q_c, h_c, η_c). We couple these variables with weights on the corners. Exact binary-product inequalities, as in (61), define $h_{\text{on}} = uh$ and $\eta_{\text{on}} = u\eta$. Nonnegative weights z_c on the corners satisfy

$$\begin{gathered}\sum_c z_c = u, \quad \sum_c q_c z_c = q, \quad \sum_c h_c z_c = h_{\text{on}}, \quad \sum_c \eta_c z_c = \eta_{\text{on}}, \\ \gamma e_{\min} \sum_c q_c h_c \eta_c z_c \le P \le \gamma e_{\max} \sum_c q_c h_c \eta_c z_c.\end{gathered} \tag{63}$$

Here $e_{\min}, e_{\max}$ bound electrical efficiency over the power range. Their use assumes $qh\eta \ge 0$, which follows from (13) for nonnegative power and positive electrical efficiency. Commitment alone does not imply $h \ge 0$: an idle unit with zero minimum power may have any head in its declared running domain. The corner box retains that domain, and bounds on h_{on} include both zero and any negative running heads. For any point in the box, multilinear interpolation weights reproduce its coordinates and the product $qh\eta$. Hence the inequalities include every physical running point. When $u = 0$, all corner weights vanish. The original efficiency and power equalities still determine the physical output.

For a tabulated turbine, we can use its head knots h_j and weights μ_j to bound power. Define $\zeta_j = \mu_j$ for a running unit and $\zeta_j = 0$ for an off unit through the following constraints:

$$\begin{gathered}0 \le \zeta_j \le \mu_j, \quad \sum_j \zeta_j = u, \quad \sum_j (h_j - h_1)\zeta_j = h_{\text{on}} - h_1 u, \\ P \le aq + \sum_j c_j \zeta_j.\end{gathered} \tag{64}$$

The coefficients c_j are chosen so that the weighted expression bounds the polynomial upper bound p^{ub}, including its variation between head knots. The constraints give $\zeta = \mu$ when $u = 1$, and $\zeta = 0$ when $u = 0$ without fixing the head of an off unit. Discharge weights support analogous bounds along the other table axis. We may add further planes during the search after verifying each over its complete domain.

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